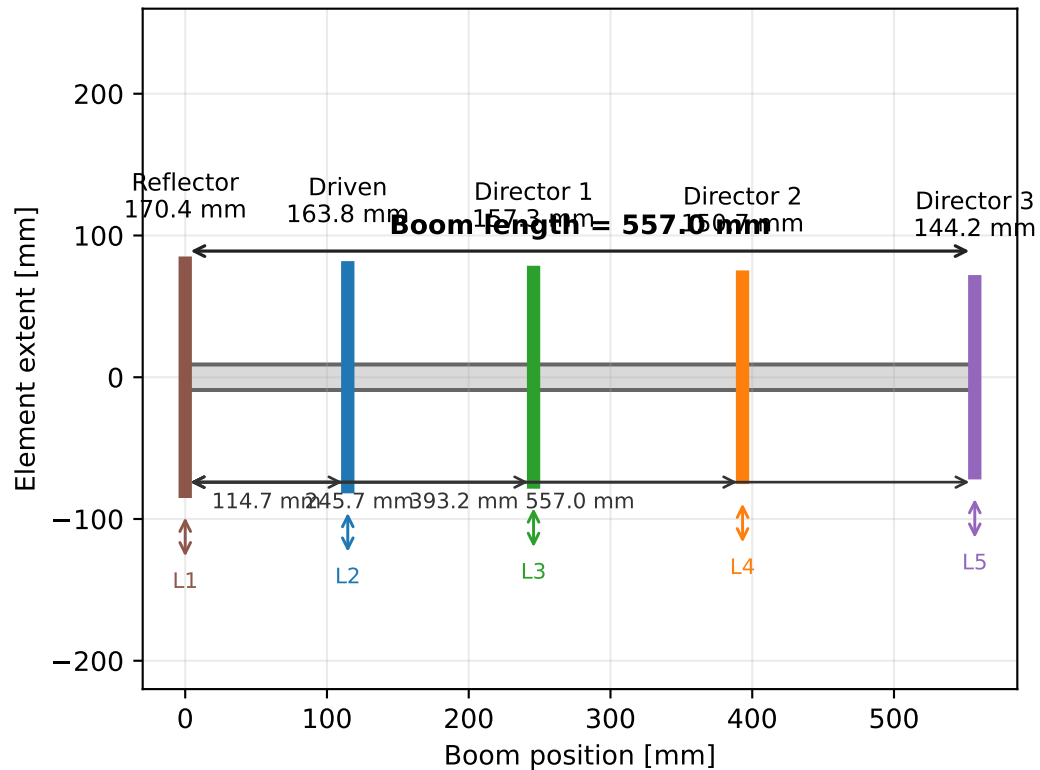


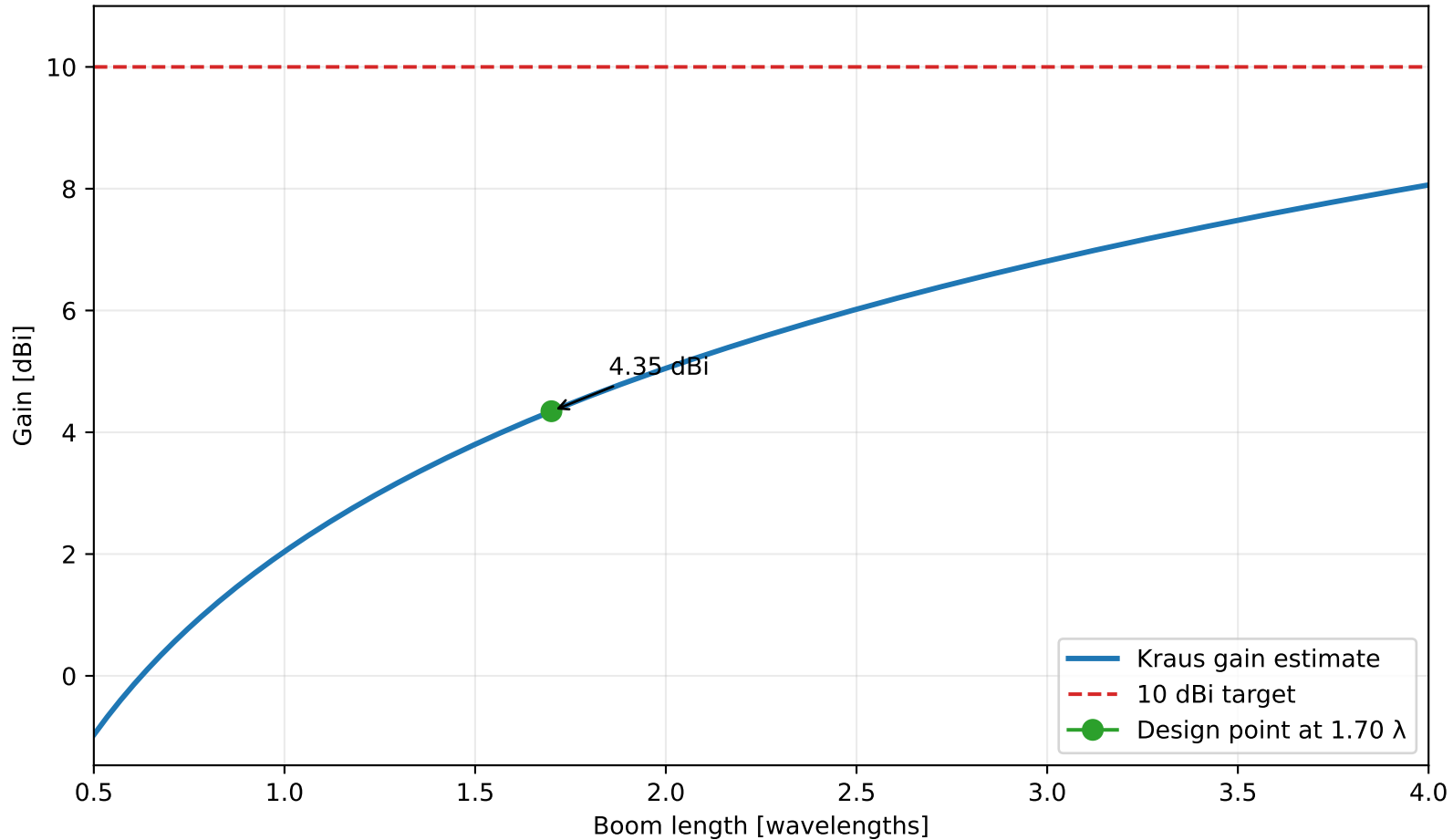
Yagi-Uda 915 MHz Design Summary

Parameter	Value
Operating frequency	915.000 MHz
Free space wavelength	327.64 mm
Reflector length	170.37 mm
Driven element length	163.82 mm
Director 1 length	157.27 mm
Director 2 length	150.72 mm
Director 3 length	144.16 mm
Reflector position	0.00 mm
Driven position	114.67 mm
Director 1 position	245.73 mm
Director 2 position	393.17 mm
Director 3 position	556.99 mm
Total boom length	556.99 mm
Driven feed impedance	25.00 ohm
VSWR into 50 ohms	2.000:1
Return loss	9.542 dB
Mismatch loss	0.512 dB
Gain, Viezbicke (design)	11.055 dBi
Gain, Kraus	5.465 dBi
Half power beamwidth	28.286 deg
Maximum pointing error	14.143 deg
Front to back ratio	20.00 dB
Front to back power ratio	100.00 x
Effective aperture	1089.18 cm ²
Link margin improvement vs isotropic	11.055 dB
Link margin improvement vs 3 dBi whip	8.055 dB
Reference minimum link margin	32.29 dB
Hairpin match length	20.478 mm

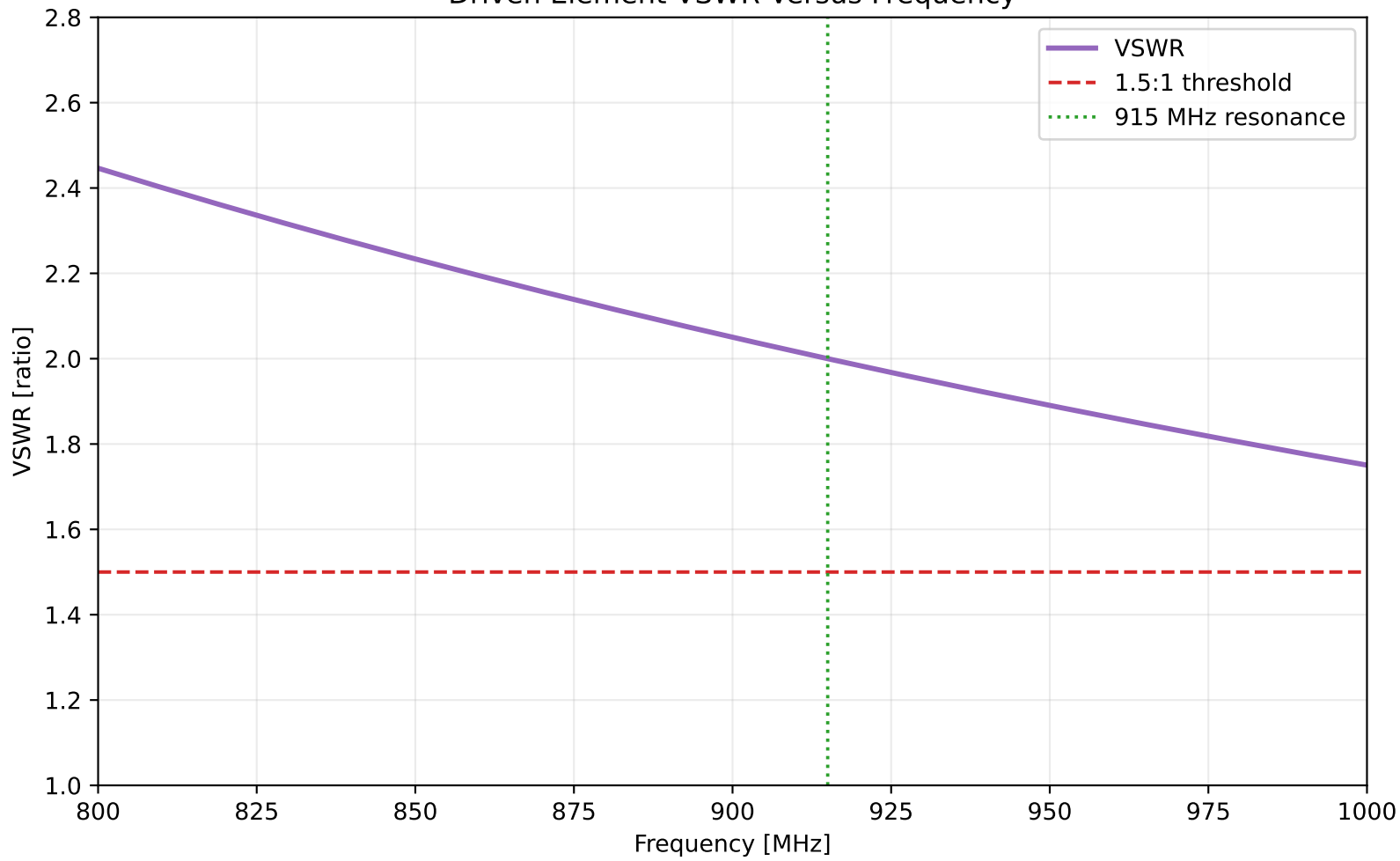
5-Element Yagi-Uda Element Layout at 915 MHz



Gain Versus Boom Length
Design average gain = 11.06 dBi



Driven Element VSWR Versus Frequency



Pointing Error Impact on Received Gain

